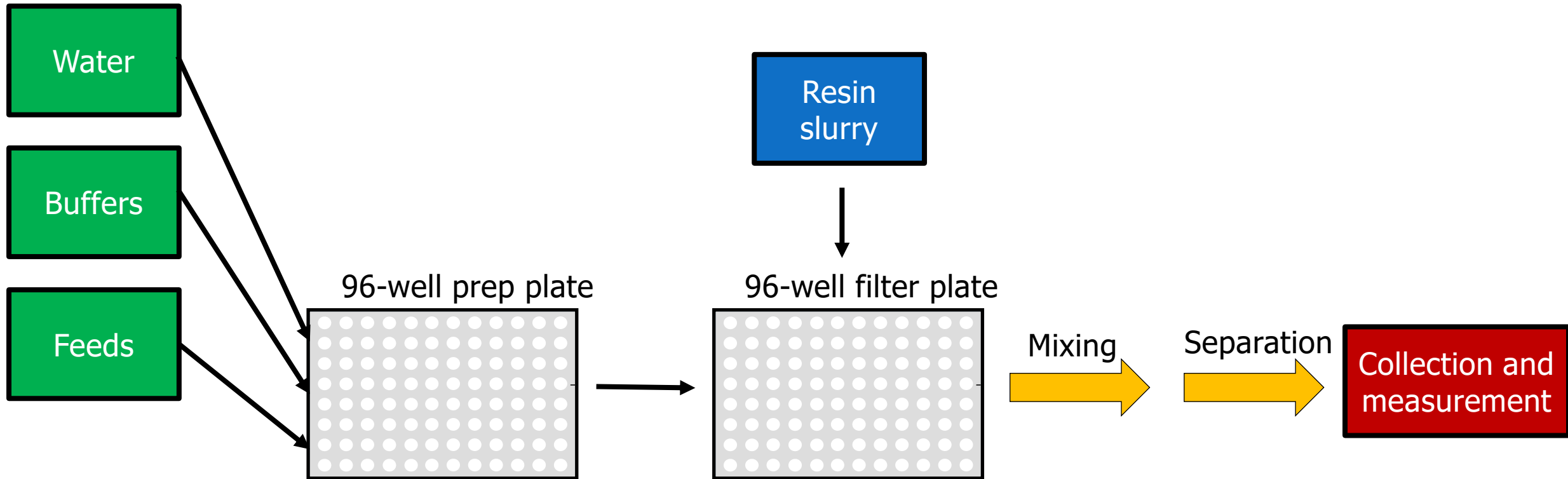


Slurry plate experimental steps



Slurry plate operations overview

Workbook prep
Complete run worklist with buffer/feed stocks
and desired experimental conditions
Iterate as necessary
Execute macros to generate pipetting scripts

Sample prep
Buffer prep
Feed prep with pH adjustment
Troughs and tubes consumables

Resin prep
Gravity settling to determine slurry %
Vigorous mixing in trough
Multichannel pipetting from trough to filter plates

Tecan screen

Equilibration x 3

Load

Strip

Sequence of steps:
Tecan pipetting into plates
Mixing with TeShake (5-60 min)
Separate liquid fraction with TeVac

Analysis
UV of load, strip plates
UPSEC of load plates
Paste data into analysis workbook

Guidance provided for analysis template



K_p analysis template:

General

K_p calculation requires q and C values:

- C is measured
- q is determined from a mass balance:
 - V_{liq} and V_{res} are entered
 - C₀ can be entered or measured

$$K_P = \frac{q}{C} = \frac{[bound]}{[liquid]}$$

$$q = \frac{V_{liq}(C_0 - C)}{V_{res}}$$

Template determines K_p values for a 96-well plate format

Cells have the following color scheme for data entry:

Empty
Required entry
Optional entry
Formula (do not modify)

K_p analysis template:

Inputs tab

Required entries
for K_p calculation:

- Resin volume
- Load volume
- Values can be entered using the uniform value cells (recommended) or as individual entries
- Entries in the uniform value cell will prepopulate the entire plate

$$K_p = \frac{q}{C}$$

$$q = \frac{V_{liq}(C_0 - C)}{V_{res}}$$

Uniform value (optional):

Buffer:	
pH:	
Buffer conc (mM):	

Salt:	
Salt/cond units:	
Salt/cond conc:	

Resin:	
Resin vol, V _{res} (μL):	50
Load vol, V _{liq} (μL):	300
Load conc, C ₀ (g/L):	

Holdup vol, V _{holdup} (μL):	
---------------------------------------	--

Resin vol, V_{res} (μL)

50	50	50	50	50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50	50	50	50	50

Load vol, V_{liq} (μL)

300	300	300	300	300	300	300	300	300	300	300	300
300	300	300	300	300	300	300	300	300	300	300	300
300	300	300	300	300	300	300	300	300	300	300	300
300	300	300	300	300	300	300	300	300	300	300	300
300	300	300	300	300	300	300	300	300	300	300	300
300	300	300	300	300	300	300	300	300	300	300	300
300	300	300	300	300	300	300	300	300	300	300	300
300	300	300	300	300	300	300	300	300	300	300	300

K_p analysis template:

Inputs tab

Optional entries:

Uniform value (optional):

Buffer:	
pH:	
Buffer conc (mM):	

Salt:	
Salt/cond units:	
Salt/cond conc:	

Resin:	
Resin vol, V _{res} (μL):	

Load vol, V _{liq} (μL):	
Load conc, C ₀ (g/L):	

Holdup vol, V _{holdup} (μL):	
---------------------------------------	--

- Other entries generally used for plate layout design and record keeping
- Load concentration, C₀, can be entered here or measured directly from the UV tab. It is not required in this tab, but a value is required for K_p determination.

$$K_P = \frac{q}{C} \qquad q = \frac{V_{liq}(C_0 - C)}{V_{res}}$$

K_p analysis template:

Inputs tab

Optional entries:

Uniform value (optional):

Buffer:	
pH:	
Buffer conc (mM):	

Salt:	
Salt/cond units:	
Salt/cond conc:	

Resin:	
Resin vol, V _{res} (μL):	50

Load vol, V _{liq} (μL):	300
Load conc, C ₀ (g/L):	

Holdup vol, V _{holdup} (μL):	35
---------------------------------------	----

Holdup volume considerations

- Ex) Assume 70% intraparticle porosity for 50 μL resin volume

Holdup vol, V_{holdup} (μL)

35	35	35	35	35	35	35	35	35	35	35	35
35	35	35	35	35	35	35	35	35	35	35	35
35	35	35	35	35	35	35	35	35	35	35	35
35	35	35	35	35	35	35	35	35	35	35	35
35	35	35	35	35	35	35	35	35	35	35	35
35	35	35	35	35	35	35	35	35	35	35	35
35	35	35	35	35	35	35	35	35	35	35	35
35	35	35	35	35	35	35	35	35	35	35	35

- Mass balance is adjusted as follows:

$$q = \frac{V_{liq}(C_0 - C) - V_{holdup}C}{V_{res}}$$

- Recommendation for a consistent approach; either never use or always use

K_p analysis template:

UV calibration and raw data tab

Standard curve for protein concentration

Required entries:

- Stock concentrations
 - Example depicts 2x dilutions
- A280
 - Template is designed to plot triplicate values
- Regression values
 - Manual entry from auto-generated curve

Optional entry:

- A320 (recommended)
 - Light scatter background correction

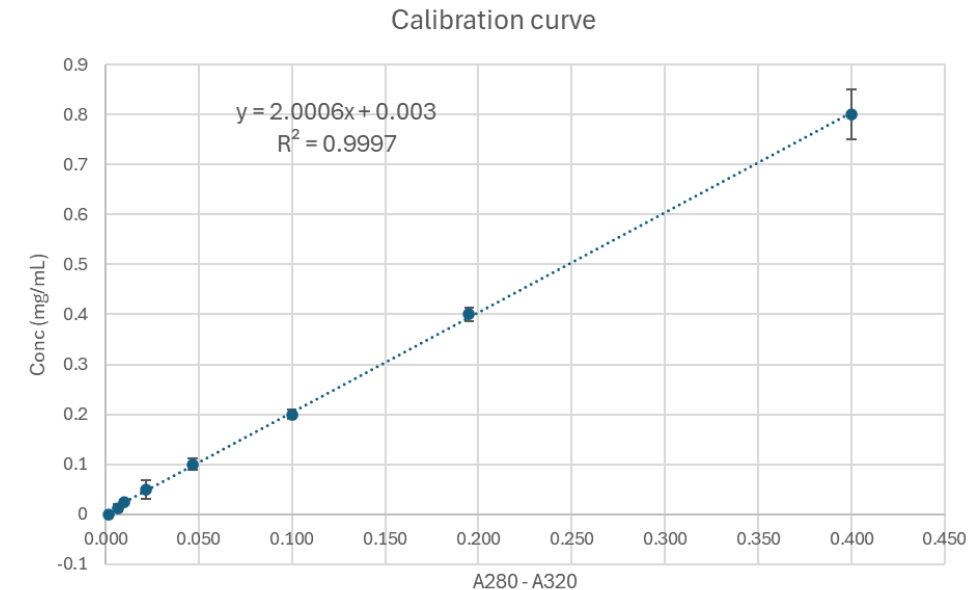
A280		
0.5	0.55	0.45
0.3	0.3	0.28
0.2	0.21	0.18
0.15	0.16	0.15
0.125	0.11	0.13
0.11	0.115	0.11
0.105	0.1	0.11
0.1	0.09	0.1

A320		
0.1	0.1	0.1
0.095	0.1	0.1
0.1	0.1	0.09
0.11	0.1	0.11
0.09	0.11	0.1
0.1	0.105	0.1
0.1	0.1	0.095
0.1	0.09	0.095

A280 - A320			Average	StDev
0.4	0.45	0.35	0.400	0.050
0.205	0.2	0.18	0.195	0.013
0.1	0.11	0.09	0.100	0.010
0.04	0.06	0.04	0.047	0.012
0.035	0	0.03	0.022	0.019
0.01	0.01	0.01	0.010	0.000
0.005	0	0.015	0.007	0.008
0	0	0.005	0.002	0.003

Stock conc (mg/mL)
0.8
0.4
0.2
0.1
0.05
0.025
0.0125
0

y = mx + b	
m	2.0006
b	0.003



K_p analysis template:

UV calibration and raw data tab

Standard curve for pathlength correction (optional)

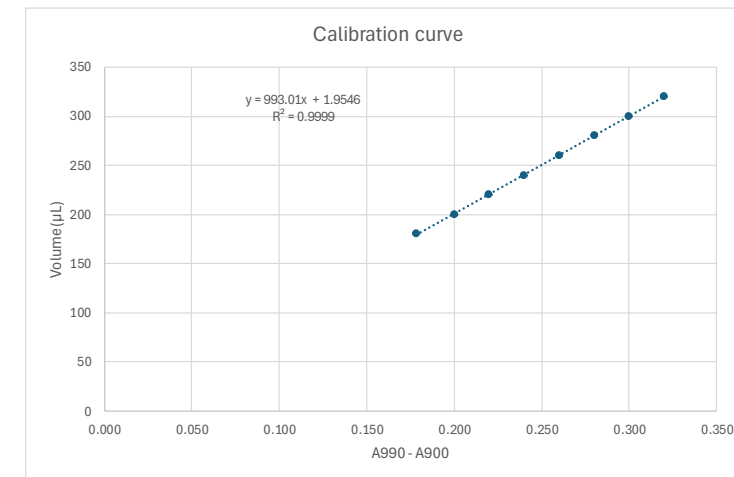
- Calibrates and measures well volume in load flowthrough only
 - Assumes initial liquid volume is correct as entered
- Additional holdup volume will be captured with this calculation
 - Recommendation to use either A990-A900 or holdup correction, but not both

$$q = \frac{(V_{liq}C_0 - V_{liq,corrected}C)}{V_{res}}$$

Standard curve entries (right side of tab)

A990	A900	A990 - A900	Average	StDev	Standard vol (μL)
0.42	0.41	0.43			320
0.4	0.39	0.41			300
0.38	0.37	0.39			280
0.36	0.35	0.37			260
0.34	0.33	0.35			240
0.32	0.31	0.33			220
0.3	0.29	0.31			200
0.28	0.27	0.285			180
0.1	0.1	0.1	0.320	0.010	
0.1	0.1	0.1	0.3	0.010	
0.1	0.1	0.1	0.280	0.010	
0.1	0.1	0.1	0.260	0.010	
0.1	0.1	0.1	0.240	0.010	
0.1	0.1	0.1	0.220	0.010	
0.1	0.1	0.1	0.200	0.010	
0.1	0.1	0.1	0.185	0.008	

Pathlength calibration	
y = mx + b	
m	993.01
b	1.9546



Corrected volume calculation (bottom right)

Corrected vol (μL)											
299.8576	279.9974	279.9974	279.9974	279.9974	279.9974	279.9974	279.9974	279.9974	279.9974	279.9974	279.9974
299.8576	299.8576	299.8576	299.8576	299.8576	299.8576	299.8576	299.8576	299.8576	299.8576	299.8576	299.8576
309.7877	309.7877	309.7877	309.7877	309.7877	309.7877	309.7877	309.7877	309.7877	309.7877	309.7877	309.7877
289.9275	289.9275	289.9275	289.9275	289.9275	289.9275	289.9275	289.9275	289.9275	289.9275	289.9275	289.9275
270.0673	270.0673	270.0673	270.0673	270.0673	270.0673	270.0673	270.0673	270.0673	270.0673	270.0673	270.0673
339.578	339.578	339.578	339.578	339.578	339.578	339.578	339.578	339.578	339.578	339.578	339.578
309.7877	309.7877	309.7877	309.7877	309.7877	309.7877	309.7877	309.7877	309.7877	309.7877	309.7877	309.7877
319.7178	319.7178	319.7178	319.7178	319.7178	319.7178	319.7178	319.7178	319.7178	319.7178	319.7178	319.7178

K_p analysis template:

UV calibration and raw data tab

Protein concentration measurements

Required entry:

- A280 from load flowthrough
 - Needed with calibration curve to calculate the liquid concentration, C

Load flowthrough

A280

0.09	0.1	0.1	0.1	0.1	0.1	0.09	0.1	0.1	0.09	0.09	0.1
0.1	0.15	0.3	0.09	0.12	0.15	0.11	0.1	0.1	0.1	0.1	0.1
0.11	0.25	0.4	0.15	0.15	0.25	0.2	0.15	0.09	0.1	0.1	0.1
0.2	0.3	0.5	0.2	0.3	0.35	0.3	0.25	0.15	0.2	0.1	0.1
0.3	0.4	0.5	0.25	0.35	0.4	0.4	0.4	0.2	0.3	0.15	0.15
0.4	0.5	0.5	0.3	0.4	0.45	0.5	0.4	0.3	0.4	0.25	0.15
0.5	0.5	0.5	0.3	0.5	0.5	0.55	0.5	0.4	0.5	0.35	0.25
0.55	0.5	0.5	0.5	0.55	0.5	0.5	0.5	0.5	0.55	0.4	0.35

Optional entries:

- A320 from load flowthrough (recommended)
- A280 (and A320) from load plate
 - Enter if direct measurement of C₀ is preferred over assumed value
- A280 (and A320) from strip flowthrough
 - Not necessary but can be helpful for troubleshooting

A320

0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1

K_p analysis template:

Results tab

Optional entries:

- Impurities 1 and 2
 - C and C₀ inputs needed from outside source
 - A standard feed value can also be used for C₀
- Target protein inputs are imported from previous tabs

Impurity 1 name:	HMW
Impurity 1 units:	ug/mL
Impurity 1 conc, C ₀ :	80

Impurity 1: HMW Units: ug/mL

80	80	80	80	80	80	80	80	80	80	80	80
80	80	80	80	80	80	80	80	80	80	80	80
80	80	80	80	80	80	80	80	80	80	80	80
80	80	80	80	80	80	80	80	80	80	80	80
80	80	80	80	80	80	80	80	80	80	80	80
80	80	80	80	80	80	80	80	80	80	80	80
80	80	80	80	80	80	80	80	80	80	80	80
80	80	80	80	80	80	80	80	80	80	80	80
80	80	80	80	80	80	80	80	80	80	80	80

C Value <>0

0.0001	0.0001	0.0001	0.001	0.001	0.001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
0.0001	0.0001	5	0.001	0.001	0.001	5	5	5	0.0001	0.0001	0.0001
0.0001	5	10	0.001	0.001	5	10	5	5	0.0001	0.0001	0.0001
0.0001	20	25	0.001	5	5	30	20	5	0.0001	0.0001	0.0001
10	30	40	0.001	10	20	40	25	15	10	0.0001	0.0001
20	40	50	10	20	25	50	50	35	20	0.0001	0.0001
30	60	80	20	30	50	80	70	60	30	20	10
50	80	80	40	60	80	85	80	65	60	40	20

K_p analysis template:

Results tab

Optional entries:

- Impurities 1 and 2
 - C and C₀ inputs needed from outside source
 - A standard feed value can also be used for C₀
- Target protein inputs are imported from previous tabs

Outputs:

- K_p and color-coded log K_p values for all cells with required entries
 - log is base 10
- Color-coded separation factor, log α, values for all cells with target protein and impurity required entries

$$\alpha = \frac{K_P \text{ Impurity}}{K_P \text{ Target protein}}$$

log K _p , target protein		Max value:	2									
2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
2.000	1.628	0.806	2.000	2.000	1.628	2.000	2.000	2.000	2.000	2.000	2.000	2.000
2.000	1.021	0.359	1.628	1.628	1.021	1.270	1.628	2.000	2.000	2.000	2.000	2.000
1.270	0.806	0.000	1.270	0.806	0.595	0.806	1.021	1.628	1.270	2.000	2.000	2.000
0.806	0.359	0.000	1.021	0.595	0.359	0.359	0.359	1.270	0.806	1.628	1.628	1.628
0.359	0.000	0.000	0.806	0.359	0.044	0.000	0.359	0.806	0.359	1.021	1.628	1.628
0.000	0.000	0.000	0.806	0.000	0.000	0.000	0.000	0.359	0.000	0.595	1.021	1.021
0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.359	0.595	0.595

log K _p , impurity 1		Max value:	2									
2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
2.000	2.000	1.954	2.000	2.000	2.000	1.954	1.954	1.954	2.000	2.000	2.000	2.000
2.000	1.954	1.623	2.000	2.000	1.954	1.623	1.954	1.954	2.000	2.000	2.000	2.000
2.000	1.255	1.121	2.000	1.954	1.954	1.000	1.255	1.954	2.000	2.000	2.000	2.000
1.623	1.000	0.778	2.000	1.623	1.255	0.778	1.121	1.415	1.623	2.000	2.000	2.000
1.255	0.778	0.556	1.623	1.255	1.121	0.556	0.556	0.887	1.255	2.000	2.000	2.000
1.000	0.301	0.000	1.255	1.000	0.556	0.000	0.000	0.301	1.000	1.255	1.623	1.623
0.556	0.000	0.000	0.778	0.301	0.000	0.000	0.000	0.141	0.301	0.778	1.255	1.255

log alpha 1												
0	0.000	0	0	0	0	0	0	0	0	0	0	0
0	0.372	1.148178	0	0	0.371507	-0.04576	-0.04576	-0.04576	0	0	0	0
0	0.933	1.264157	0.371507	0.371507	0.933461	0.353423	0.32575	-0.04576	0	0	0	0
0.730174	0.449	1.120574	0.730174	1.148178	1.359472	0.193936	0.234491	0.32575	0.730174	0	0	0
0.817185	0.641	0.778151	0.979218	1.028479	0.89618	0.419059	0.761482	0.145147	0.817185	0.371507	0.371507	0.371507
0.89618	0.778	0.556303	0.817185	0.89618	1.076266	0.556303	0.19721	0.081232	0.89618	0.979218	0.371507	0.371507
1	0.301	0	0.449208	1	0.556303	0	0	-0.05806	1	0.660502	0.602468	0.602468
0.556303	0.000	0	0.778151	0.30103	0	0	0	0.141329	0.30103	0.419059	0.660502	0.660502

K_p analysis template:

Results tab

Optional entries:

- Impurities 1 and 2
 - C and C₀ inputs needed from outside source
 - A standard feed value can also be used for C₀
- Target protein inputs are imported from previous tabs

Outputs:

- K_p and color-coded log K_p values for all cells with required entries
 - log is base 10
- Color-coded separation factor, log α, values for all cells with target protein and impurity required entries

$$\alpha = \frac{K_{P \text{ Impurity}}}{K_{P \text{ Target protein}}}$$

Key considerations:

- C cannot be zero since it is the denominator for calculating K_p
 - A convenient workaround when it arises is to substitute a very small value (i.e., 0.0001)
- Max values for log K_p related to assay sensitivities should be considered
 - 2 is a common max value for UV measurements
 - Higher values up to 3 or 4 can be considered for more sensitive assays like HCP ELISA
- Impurities should be entered as concentrations not related to the product (i.e., ng/mL instead of % or ppm)

log K _p , target protein	Max value:	2
-------------------------------------	------------	---

log K _p , impurity 1	Max value:	2
---------------------------------	------------	---

K_P analysis template:

Importance of C₀ and C measurements

Problematic scenarios:

- At conditions of weak binding, C₀ ≈ C
 - If C₀ is too high, q never approaches zero
 - If C₀ is too low, q will have negative values
- At conditions of strong binding, C ≈ 0
 - If C₀ is too high, q is artificially elevated
 - If C₀ is too low, q is artificially lowered

$$K_P = \frac{q}{C}$$

$$q = \frac{V_{liq}(C_0 - C)}{V_{res}}$$

Common sources of error and mitigations:

- Feed stock is improperly measured or diluted
 - Measure C₀ rather than assuming a value
 - Include holdup volumes
 - Recalibrate standard curve
- Well volumes are inconsistent
 - Use A990-A900
- Light scatter between different samples (salts) is inconsistent
 - Subtract A320 from A280

A well-designed screen should have several points expected to be at the min and max of the detection range with most points in between

- All results should be reviewed with adjustments made to C₀ or absorbance data, if needed
- ***Consistency in approach is critical***

K_P analysis template:

Importance of C₀ and C measurements

Template prioritizes C₀ measurement entries as follows:

- 1) Uniform value (Inputs tab)
 - Simplest approach
- 2) Individually entered values (Inputs tab)
- 3) Measured values (UV and raw data tab)
 - Leave C₀ Input tab entries blank if using this approach
 - Values for load flowthrough max and load measured average are also recorded (UV and raw data tab)
 - These values may be substituted for the Uniform value on the entry tab when assumed C₀ values are incorrect

Load flowthrough

Conc, C (mg/mL) Measured max: 0.903

$$K_P = \frac{q}{C}$$

$$q = \frac{V_{liq}(C_0 - C)}{V_{res}}$$

1)

Load conc, C ₀ (g/L):	0.833
----------------------------------	-------

2)

Load conc, C ₀ (g/L)											
0.833	0.833	0.833	0.833	0.5	0.5	0.5	0.5	1	1	1	1
0.833	0.833	0.833	0.833	0.5	0.5	0.5	0.5	1	1	1	1
0.833	0.833	0.833	0.833	0.5	0.5	0.5	0.5	1	1	1	1
0.833	0.833	0.833	0.833	0.5	0.5	0.5	0.5	1	1	1	1
0.833	0.833	0.833	0.833	0.5	0.5	0.5	0.5	1	1	1	1
0.833	0.833	0.833	0.833	0.5	0.5	0.5	0.5	1	1	1	1
0.833	0.833	0.833	0.833	0.5	0.5	0.5	0.5	1	1	1	1
0.833	0.833	0.833	0.833	0.5	0.5	0.5	0.5	1	1	1	1

3)

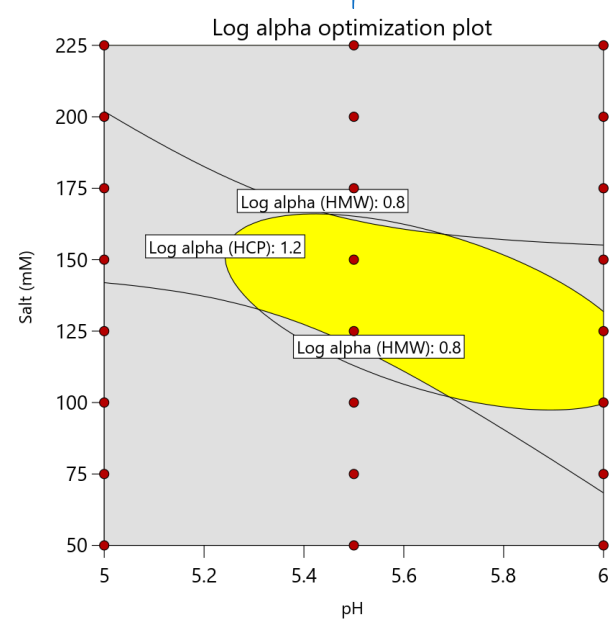
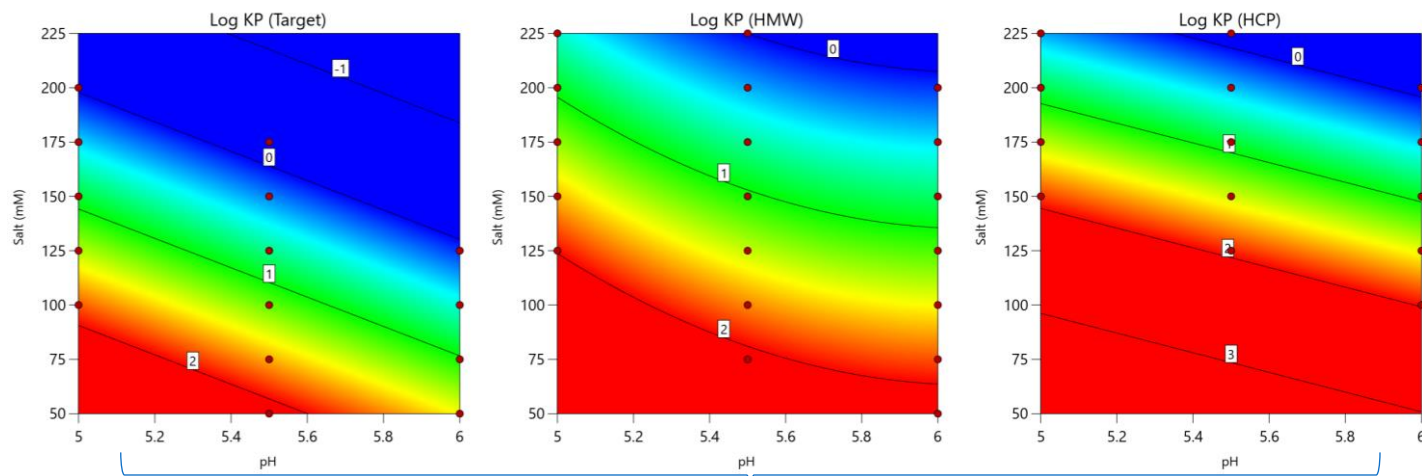
Load plate initial (optional)											
A280											
0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52
0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51
0.49	0.49	0.49	0.49	0.49	0.49	0.49	0.49	0.49	0.49	0.49	0.49
0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55
0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51

Conc, C ₀ (mg/mL)	Measured average:											
0.843	0.843	0.843	0.843	0.848	0.843	0.843	0.843	0.843	0.843	0.843	0.843	0.843
0.803	0.803	0.803	0.803	0.803	0.803	0.803	0.803	0.803	0.803	0.803	0.803	0.803
0.823	0.823	0.823	0.823	0.823	0.823	0.823	0.823	0.823	0.823	0.823	0.823	0.823
0.783	0.783	0.783	0.783	0.783	0.783	0.783	0.783	0.783	0.783	0.783	0.783	0.783
1.003	1.003	1.003	1.003	1.003	1.003	1.003	1.003	1.003	1.003	1.003	1.003	1.003
0.903	0.903	0.903	0.903	0.903	0.903	0.903	0.903	0.903	0.903	0.903	0.903	0.903
0.803	0.803	0.803	0.803	0.803	0.803	0.803	0.803	0.803	0.803	0.803	0.803	0.803
0.823	0.823	0.823	0.823	0.823	0.823	0.823	0.823	0.823	0.823	0.823	0.823	0.823

Visually plotting results example:

Comparison of surface plots and raw data

Log K_p and alpha optimization surface plots (separate software)



Sanity check with log K_p data

	log K_p , target			log K_p , HMW			log K_p , HCP		
	5	5.5	6	5	5.5	6	5	5.5	6
225	0.00	0.00	0.00	0.56	0.00	0.00	0.00	0.00	0.00
200	0.00	0.00	0.00	1.00	0.30	0.00	0.78	0.78	0.00
175	0.36	0.00	0.00	1.26	0.78	0.56	1.26	0.78	0.30
150	0.81	0.36	0.00	1.62	1.00	0.78	2.00	1.73	0.78
125	1.27	0.81	0.00	2.00	1.26	1.12	2.00	2.00	1.26
100	2.00	1.02	0.36	2.00	1.95	1.62	2.00	2.00	2.00
75	2.00	1.63	0.81	2.00	2.00	1.95	2.00	2.00	2.00
50	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00

- Yellow highlights identified from surface plots
 - Aligns with data for bind-and-elute
 - High salt less optimal for HMW separation
- Orange highlights also look feasible